

Laser Particle Counter Calibration

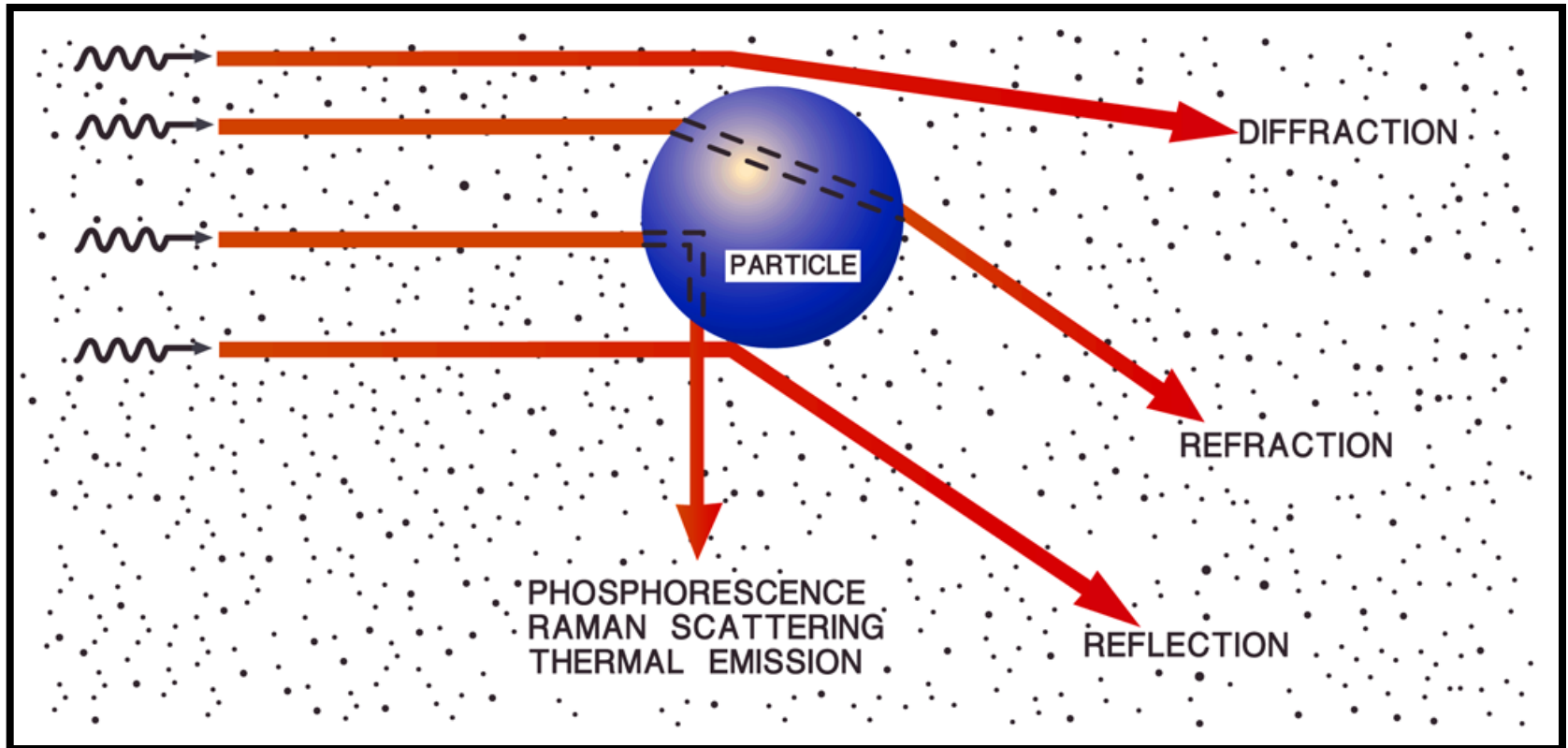
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Agenda

- Basic principles of operation
- Risk points for functionality and calibration
- History of calibration
- ISO21501-4

Light Scattering

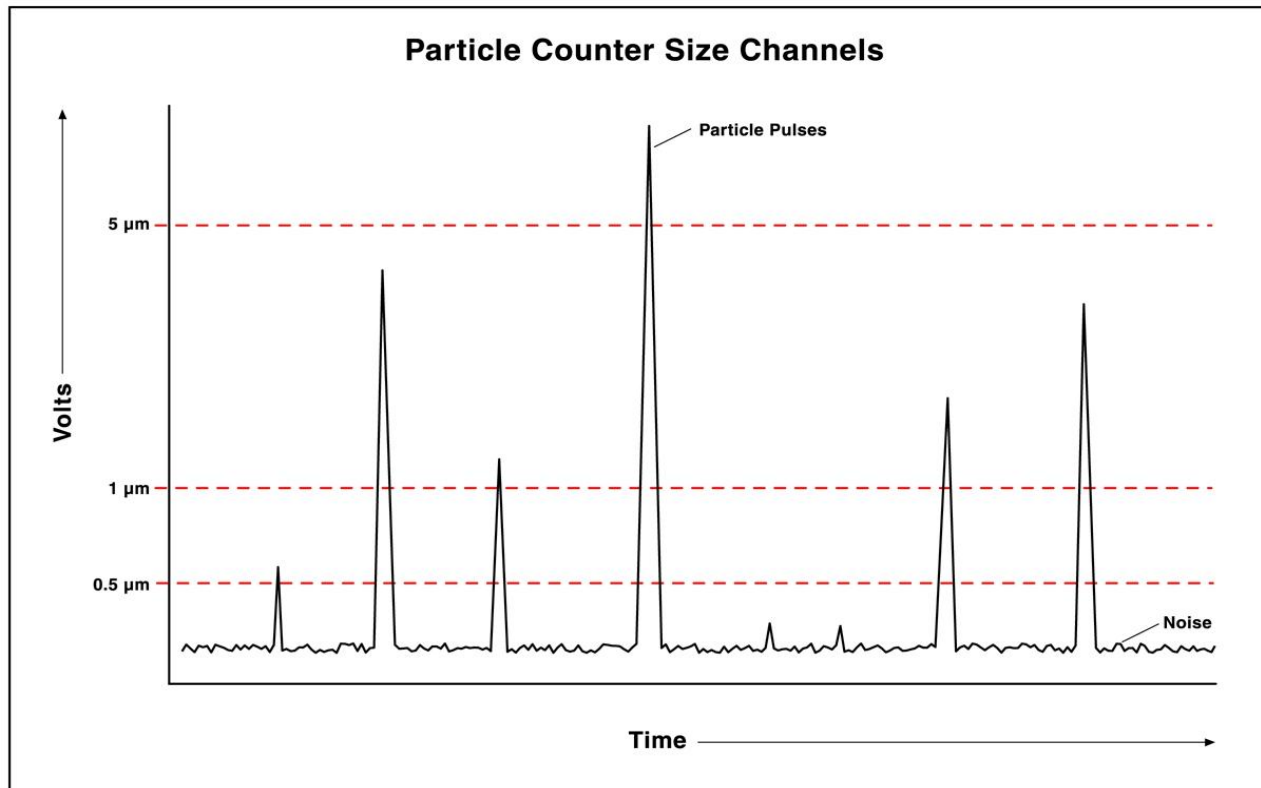


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Thresholds

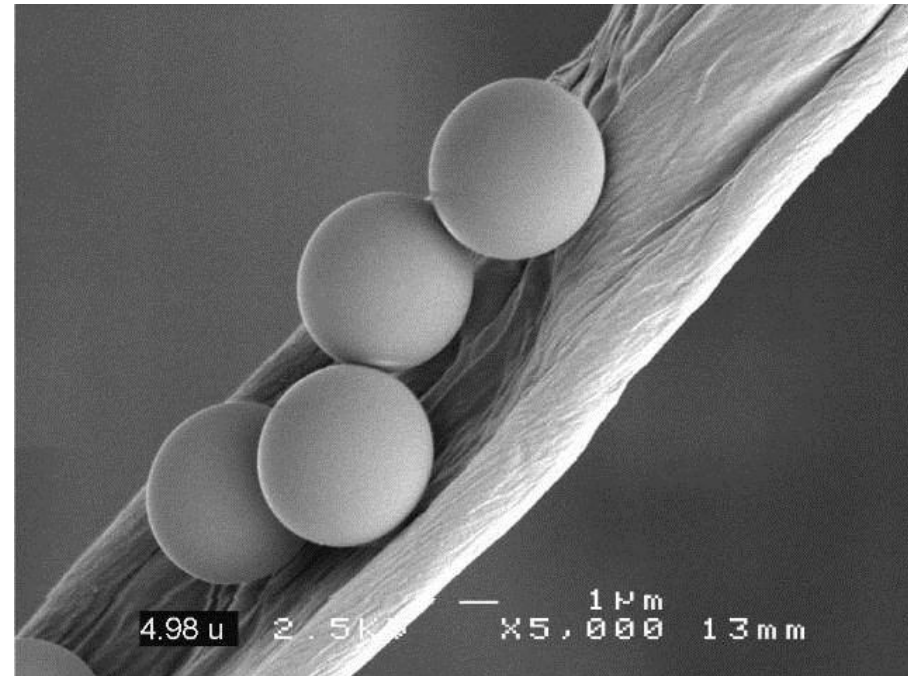
- Thresholds designed for specific particle sizes
 - Each particle pulse compares to threshold and put into size channel



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Calibration Materials

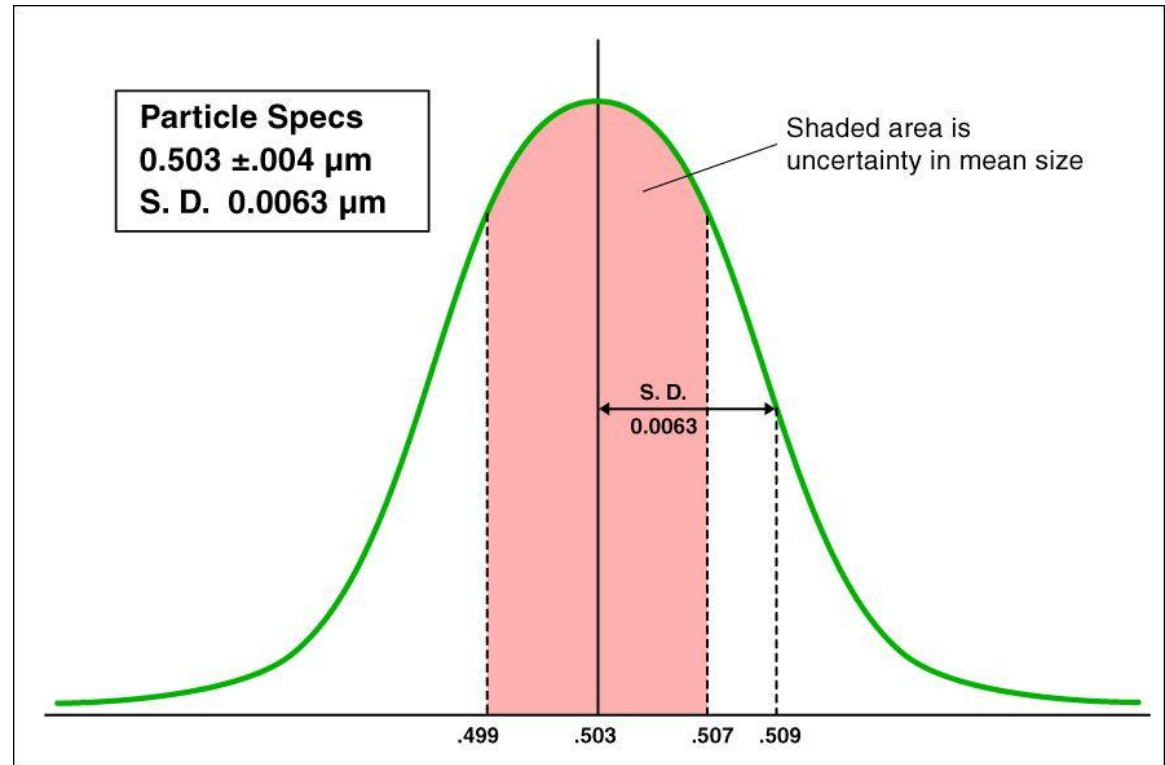
- Polystyrene latex spheres (PSL's)



4.98 µm PSLs on a cotton fiber

Polystyrene Latex Spheres

- PSL Sizing
 - $0.503\ \mu\text{m}$
 - $\pm 0.004\ \mu\text{m}$
 - $\sigma: 0.0063\ \mu\text{m}$
 - Concentration:
 1×10^9 per mL



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Air Particle Counter Calibration

- No traceable concentration standard exists
- Calibration is therefore about the sizing accuracy of an instrument
- Particle Standards *are* available

ISO 21501-4

- International standard ISO 21501-4 sets guidelines for the calibration and verification of optical particle counters (OPC) for airborne particles.
- **Objective**: To provide a calibration and verification method for airborne particle counters, so as to minimize the inaccuracy in the measurement result by a counter, as well as the differences in the results measured by different instruments

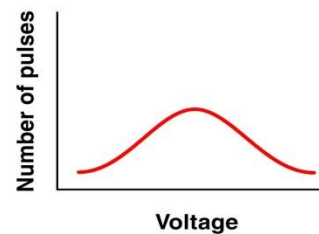
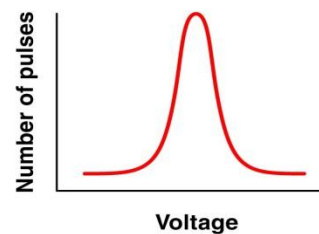
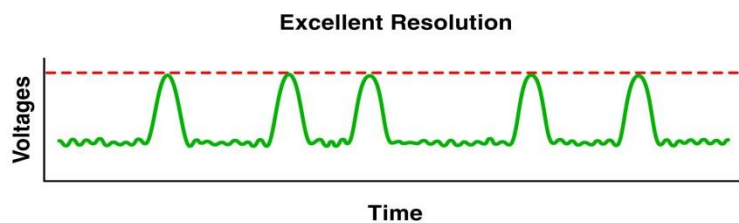
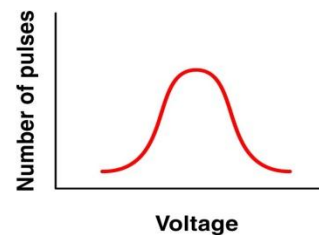
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ISO 21501-4 Calibration Procedure

- Use a Pulse Height Analyzer (PHA) to set size channels
- Set threshold to median of Pulse Height Distribution (PHD)
- Verify resolution meets requirement
- Verify counting efficiency of first channel and at 1.5-2.0 x minimum size using reference unit with known sensitivity
- Zero Count test

PHA?

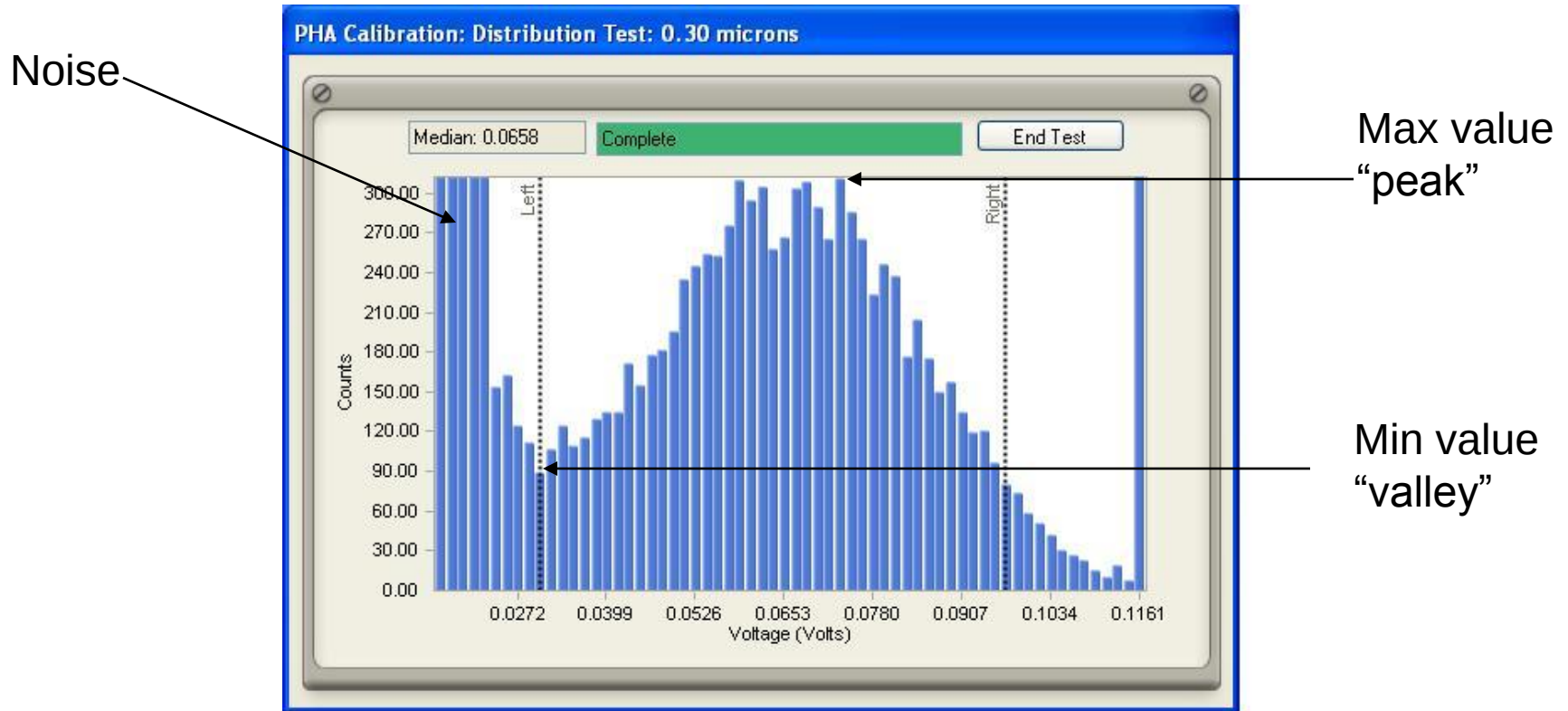


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Actual PHD

Max must be 2x Min on Most Sensitive Channel



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A typical Calibration Rig:



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ISO 21501-4

(light scattering Air-borne particle counter)

- **Objective**

- To provide a calibration and verification method for airborne particle counters, so as to minimize the inaccuracy in the measurement result by a counter, as well as the differences in the results measured by different instruments.

- **Calibration Requirements**

• Sampling flow rate	≈	volumetric flow rate $\pm 5\%$
• Counting efficiency @ particle size close to the minimum detectable size (1 st channel)	≈	50% $\pm 20\%$
• Counting efficiency @ particle size of 1.5 to 2 times the minimum detectable size	≈	100% $\pm 10\%$
• Particle size setting error	≤	10%
• Instrument resolution @ the particle size specified by the counter manufacturer	≤	15%
• Zero count test	≤	1 count / 5 minutes
• Maximum particle number concentration	≈	as specified by the manufacturer
• Coincidence loss @ the maximum particle number concentration	≤	10%
• Sampling time	≈	preset value (100% $\pm 1\%$)
• Response rate	≤	0.5%
• Calibration interval	≤	1 year

- **Test Method**

- Calibration particle: PSL particles traceable to an international standard with the standard uncertainty $\leq 2.5\%$
- Sizing calibrated with a PHA @ median value for each size channel
- Pulse height distribution of PSL particle signals with noise must increase by 50% from the minimum to the peak
- Counting Efficiency Test must be verified with a reference instrument (e.g., CPC) that has smaller minimum detectable size
- Test sample particle concentration $\leq 25\%$ of the maximum allowed concentration
- False Count Rate: Observed data fitted with Poisson distribution with a 95% upper confidence limit



Testing Flow Rate

- With a flow control system (pump, blower)
 - Comparison to certified reference flow meter
 - Air – $\pm 5\%$ of nominal (marketing specification)
- Without a flow control system
 - UUT is set with certified flow system
 - Manufacturer specifies the allowable limits of its flow rate

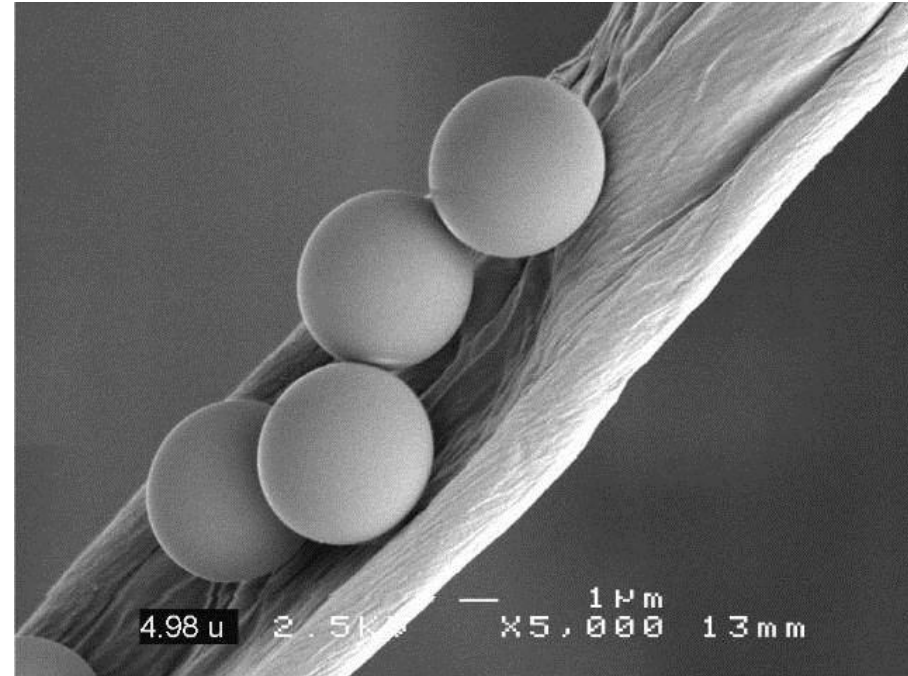
Calibration Particles

- Calibration particles
 - “Mono-disperse spherical particle with a known mean particle size, e.g. polystyrene latex (PSL) particle, that is traceable to an international standard of length, and where the standard uncertainty of the mean particle size is equal to or less than $\pm 2.5\%$.”
 - Typical size ranges
 - Parts 2 & 4 ranges 0.1 – 10.0 μm
 - Part 3 ranges 1.0 – 100 μm
- Polystyrene latex spheres
 - Available in sizes ranging from 0.021 μm and up
 - Dry and liquid varieties
 - Large sizes ($> 5 \mu\text{m}$) in the dry variety are problematic due to an uncertainty $> \pm 2.5\%$
 - NIST and JIS are the common traceable, international standards of length
 - Slight discrepancies exist in actual size between JSR/Clintex and Duke
 - Lot to lot differences exist
 - Affected by age, dilution, handling...
 - Limited shelf life
 - ‘Beginning’ of uncertainty

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Calibration Materials

- Polystyrene latex spheres (PSL's)



4.98 μm PSLs on a cotton fiber

**N.B. – These cost \$500-
\$1000 per bottle!**

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The Standard ISO-21501

- What is the minimum information necessary in a compliant calibration report?

Test Report - The Minimum Required:

- Date of calibration
- Calibration particle sizes
- Flow rate
- Size resolution (with the particle size used)
- Counting efficiency
- Voltage limit or channel of an internal pulse height analyzer (PHA)
- Verification of size setting
- False count rate in aerosol particle counters

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The Calibration Process

- A yearly calibration interval is recommended
- As Found verification
- ISO 21501-4 calibration
- What do I do if the 'As Found' test shows significant error?



5475 Airport Boulevard, Boulder, Colorado 80301-2339
303.443.7100 1.800.238.1801 Fax: 303.546.7380
Customer Service Center 1.877.475.3317
Instrument Service and Support 1.800.557.6363

Certificate of Calibration

Date: Jan-21-2010 16:37:22

Customer Name:

Service Notes:

None

RMA Number:

Unit Under Test

Instrument: LasairIII_5100

Serial Number:

Calibration Due: Jan-21-2011

Environmental

Temperature: 22.94 Deg C

Relative Humidity: 14.24%

Certification

Particle Measuring Systems certifies that the instrument listed above meets or exceeds all published specifications. It has been calibrated using equipment and standards whose accuracies are traceable to the USA National Institute of Standards and Technology (NIST) and/or Japanese Industrial Standards (JIS). Our calibration processes satisfy BS EN ISO9001:2000 requirements. This instrument was calibrated in accordance with the Particle Measuring Systems quality system documentation and meets the requirements of ISO 21501-4. This certificate may not be reproduced, except in full, without written consent from Particle Measuring Systems.

Calibration Standards

Procedure Used: 1000013097 Rev D

Instrument Condition

☒ In-Tolerance

☐ Out-of-Tolerance

☐ Operational Failure

Particles Used

<u>Particle Size (micron)</u>	<u>Standard Deviation</u>	<u>Lot Number</u>	<u>Expiration Date</u>
0.498	0.008	35455	Jan-31-2011
0.994	0.010	35018	Oct-31-2010
1.998	0.020	35466	Jul-31-2012
5.003	0.050	35825	Oct-31-2012

Flow Meter Instrument

Serial Number: 119638

Calibration Due: Sep-01-2010

Reference Instrument

Instrument: LasairII-110

Serial Number: 58574

Calibration Due: Apr-08-2010

Performed by

Technician Name: Brian Muldoon

Calibration Data Record

Date: Jan-21-2010 16:37:22

Customer Name:
Service Notes:

None

RMA Number:

Unit Under Test

Instrument: LasairIII_5100
Serial Number:

Laser Serial Number: No Change
Calibration Due: Jan-21-2011

Environmental

Temperature: 22.94 Deg C

Relative Humidity: 14.24%

Threshold Information

<u>Channel Size (micron)</u>	<u>Measured Threshold/Gain</u>	<u>Measurement Type</u>
Ch 1: 0.50	01744 / 0	Distribution
Ch 2: 1.00	06217 / 0	Distribution
Ch 3: 2.00	00776 / 1	Distribution
Ch 4: 5.00	04468 / 1	Distribution
Ch 5: 10.00	01080 / 2	Extrapolate
Ch 6: 25.00	05129 / 2	Extrapolate

Calibration Results

<u>Parameter</u>	<u>Nominal Size (micron)</u>	<u>Allowed Range</u>	<u>Results</u>
Flow Rate	n/a	97.5 to 102.5 L/min	100.7 L/min
Ch 1 Size Error	0.50	-5.0% to 5.0%	0.784%
Ch 2 Size Error	1.00	-5.0% to 5.0%	0.463%
Ch 3 Size Error	2.00	-5.0% to 5.0%	0.750%
Ch 4 Size Error	5.00	-5.0% to 5.0%	0.711%
Ch 5 Size Error	10.00	-5.0% to 5.0%	1.612%
Ch 6 Size Error	25.00	-5.0% to 5.0%	1.612%
Resolution	1.00	< 15.0%	12.67%
CE(%)	0.50	30.0% to 70.0%	54.53%
CE(%)	1.00	90.0% to 110.0%	103.28%
Zero Count	n/a	< 2 Cnt/m ³ (*1)	1 Cnt in 25.0 min

Reference Characteristics

CE(%) @ 0.498 109.10% CE(%) @ 0.994 102.00%

General Information

Firmware Version: B2B53624 1.9 Software Version: 1.60 Jun-23-2009

Notes

Note *1: Process staged intervals to achieve 95% UCL for stated concentration



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303.443.7100 1.800.238.1801 Fax: 303.546.7380
Customer Service Center 1.877.475.3317
Instrument Service and Support 1.800.557.6363

Verification Data Record

Date: Jan-21-2010 11:37:11

Customer Name:
Service Notes:

Nune

RMA Number:

Unit Under Test

Instrument: LasairIII_5100
Serial Number:

Laser Serial Number: No Change

Environmental

Temperature: 22.81 Deg C

Relative Humidity: 13.24%

Instrument Condition Received

☒ In-Tolerance

☐ Out-of-Tolerance

☐ Operational Failure

Threshold Information

<u>Channel Size (micron)</u>	<u>Measured Threshold/Gain</u>	<u>Received Threshold/Gain</u>	<u>Measurement Type</u>
Ch 1: 0.50	01807 / 0	01831 / 0	Distribution
Ch 2: 1.00	06486 / 0	06544 / 0	Distribution
Ch 3: 2.00	00798 / 1	00797 / 1	Distribution
Ch 4: 5.00	04685 / 1	04882 / 1	Distribution
Ch 5: 10.00	01134 / 2	01177 / 2	Extrapolate
Ch 6: 25.00	05443 / 2	05685 / 2	Extrapolate

Verification Results

<u>Parameter</u>	<u>Allowed Range</u>	<u>Results</u>	<u>Pass/Fail/Skipped</u>
Flow Rate	95.0 to 105.0 L/min	96.8 L/min	Pass
Pre Zero Count	< 50 Cnt/m ³ (*1)	4 Cnt in 2.0 min	Pass
Size Error @ 0.50	-10.0% to 10.0%	-0.72%	Pass
Size Error @ 1.00	-10.0% to 10.0%	-0.46%	Pass
Size Error @ 2.00	-10.0% to 10.0%	0.10%	Pass
Size Error @ 5.00	-10.0% to 10.0%	-2.36%	Pass
Size Error @ 10.00	-10.0% to 10.0%	-2.45%	Pass
Size Error @ 25.00	-10.0% to 10.0%	-2.34%	Pass
Resolution @ 1.00	< 15.0%	11.50%	Pass
CE(%) @ 0.50	n/a	52.51%	n/a
CE(%) @ 1.00	90.0% to 110.0%	101.16%	Pass

Note *1: Process staged intervals to achieve 95% UCL for stated concentration

General Information

Firmware Version: B2B53624 1.9

Software Version: 1.60 Jun-23-2009

Performed by

Technician Name: Brian Muldoon

Summary

- Standards exist for the purpose of setting guidelines and creating an equal playing field for particle counter comparison
- Key concepts
 - Theory of operation
 - Mie theory allows us to calculate the relative amount of scattered light of different sized PSL's
 - Calibration procedure
 - Different thresholds correspond to different voltages
 - Measuring this shift in threshold allows calculating sizing error
 - Size resolution
 - Sizing error
 - Counting efficiency
 - Flow rate
 - Voltage limit or channel of an internal PHA
 - Verification of size setting
 - False count rate

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That Being Said...

- In the final paragraph of test methods ISO 21501-(all) states that a calibration includes at least:
 - Size calibration
 - Size resolution
 - Counting efficiency
- AND
- Sampling Volume Uncertainty